**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 146 · Lesson 146 · Network Threats: Attack Paths and Consequences

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. When packets arrive out of order, what can TCP do?

Q2. Which layer in TCP/IP handles IP routing?

Q3. What data structure follows FIFO?

Q4. What is the result of an AND gate with inputs 1 and 0?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

phishing — Social engineering to obtain information or actions through deception.

malware — Software designed to perform harmful or unauthorised actions.

DDoS — Distributed denial-of-service intended to disrupt availability.

CIA triad — Confidentiality, integrity and availability.

## New learning

Entry point — Suspicious link, exposed service or weak account.

Threat action — Credential theft, malware execution or traffic flooding.

Effect on service — Compromised confidentiality, integrity or availability.

## Worked example

A pupil receives a fake school login link and submits a password. Explain the attack.

1. Fake login email — Identify the facts and make an initial prediction.

2. Credentials submitted — Explain the mechanism or trace the intermediate stage.

3. Account access at risk — Phishing tricks the pupil into visiting an attacker-controlled form. The secret is disclosed, then the attacker may attempt unauthorised account access. Awareness and phishing-resistant MFA help, while a firewall alone does not prevent users entering secrets into fake sites.

# We Do · Practise with guidance

A public learning site stops responding during a flood of requests from many infected machines. Name the threat and one defence.

Botnet generates traffic

Identify the facts and make an initial prediction.

Server/link overloaded

Explain the mechanism or trace the intermediate stage.

Mitigate upstream

A distributed denial-of-service (DDoS) attack aims to exhaust bandwidth or service resources. Upstream rate-limiting, DDoS protection and filtering may reduce impact; encrypting the website does not stop resource saturation.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain how phishing differs from malware and give one example of each. [6]

### Question 2

A school site is reachable but pupils' account details are exposed. Identify the affected security property and two controls. [6]

### Question 3

A network suffers DDoS and suspected password theft simultaneously. Prioritise two distinct mitigations and justify why a single firewall rule is inadequate. [8]

## Exit Ticket · Answer before checking

Encrypting a site with HTTPS is sufficient to prevent a DDoS attack.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Use sequence information to reassemble the byte stream correctly.

Q2. The internet layer.

Q3. A queue.

Q4. 0.

### You Do pages · Independent questions

Q1. Phishing deceives a user into divulging information or performing an action [3]; malware is malicious software with a damaging/unauthorised function, e.g. ransomware [3].

Q2. Confidentiality breached [2]; e.g. MFA/least privilege/encrypted connections, secure authentication, anti-phishing education or monitoring linked to attack [2+2].

Q3. DDoS upstream mitigation [2], account containment/MFA/reset [2], differing threat mechanism [2], defence-in-depth explanation [2].

### Exit Ticket · Model answer

False. Encryption protects traffic confidentiality/integrity in transit, not server capacity.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding